Vector-Valued Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Motion on a circle and centripetal acceleration**

A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).

$\vec{v}(t) = \langle -3\sin t, 3\cos t \rangle$. $\vec{a}(t) = \langle -3\cos t, -3\sin t \rangle = -\vec{r}(t)$. Since $\vec{a}(t)$ is exactly -1 times $\vec{r}(t)$, it points directly toward the origin (centripetal acceleration).

2 **[Foundational]** • **Target: Projectile motion as a vector-valued function**

A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).

$\vec{v}(t) = \langle 20, 30 - 9.8t \rangle$. Maximum height occurs when $30 - 9.8t = 0 \Rightarrow t = \frac{30}{9.8} \approx 3.06$ s.

3 **[Foundational]** • **Target: Position, velocity, and acceleration vectors**

A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

$\vec{v}(t) = \langle 2t, 3t^2 \rangle$. $\vec{a}(t) = \langle 2, 6t \rangle$.

4 **[Foundational]** • **Target: Position, velocity, and acceleration vectors**

For the same particle, find the speed at $t = 2$.

$\vec{v}(2) = \langle 4, 12 \rangle$. Speed $= |\vec{v}(2)| = \sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$.

5 **[Foundational]** • **Target: Integrating vector-valued functions**

Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

$\vec{r}(t) = \langle t^3 + C_1, t^2 + C_2 \rangle$. Using $\vec{r}(0) = \langle 1, 4 \rangle$: $C_1 = 1$, $C_2 = 4$. $\vec{r}(t) = \langle t^3 + 1, t^2 + 4 \rangle$.

6 **[Foundational]** • **Target: Total distance traveled**

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

Speed $= |\vec{v}(t)| = \sqrt{\cos^2 t + \sin^2 t} = 1$ (constant). Distance $= \int_0^{\pi} 1 \, dt = \pi$.

7 **[Foundational]** • **Target: Vectors and parametric motion combined**

A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

$\vec{v}(t) = \langle e^t, -e^{-t} \rangle$. At $t = 0$: $\vec{v}(0) = \langle 1, -1 \rangle$. Speed $= \sqrt{1^2 + (-1)^2} = \sqrt{2}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Vectors and parametric motion combined**

For the particle above, find the magnitude of the acceleration vector at $t = 0$.

$\vec{a}(t) = \langle e^t, e^{-t} \rangle$. At $t = 0$: $\vec{a}(0) = \langle 1, 1 \rangle$. $|\vec{a}(0)| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

9 **[Intermediate]** • **Target: Unit tangent vectors**

For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

$\vec{v}(t) = \langle 1, 2t \rangle$; at $t = 1$: $\vec{v}(1) = \langle 1, 2 \rangle$, $|\vec{v}(1)| = \sqrt{1 + 4} = \sqrt{5}$. $\vec{T}(1) = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$.

10 **[Intermediate]** • **Target: Unit tangent vectors**

Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.

Dividing any nonzero vector by its own magnitude produces a vector with magnitude $\frac{|\vec{v}(t)|}{|\vec{v}(t)|} = 1$ by definition — this is the standard construction of a unit vector in the same direction.

11 **[Intermediate]** • **Target: Motion on a circle and centripetal acceleration**

For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.

$\vec{v}(t) \cdot \vec{r}(t) = (-3\sin t)(3\cos t) + (3\cos t)(3\sin t) = -9\sin t\cos t + 9\sin t\cos t = 0$. This confirms the velocity vector is always perpendicular to the position vector — consistent with circular motion, where velocity is tangent to the circle.

12 **[Intermediate]** • **Target: Displacement vs. distance for vector motion**

A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

$\vec{r}(0) = \langle 0, 0 \rangle$; $\vec{r}(2) = \langle 4, 2 \rangle$. Displacement $= \vec{r}(2) - \vec{r}(0) = \langle 4, 2 \rangle$.

13 **[Intermediate]** • **Target: Displacement vs. distance for vector motion**

Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

$\vec{v}(t) = \langle 2t, 1 \rangle$. Distance $= \int_0^2 |\vec{v}(t)|\, dt = \int_0^2 \sqrt{(2t)^2 + 1^2}\, dt = \int_0^2 \sqrt{4t^2 + 1}\, dt$.

14 **[Intermediate]** • **Target: Vector operations review**

Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.

$2\vec{u} = \langle 6, -4 \rangle$; $3\vec{w} = \langle -3, 12 \rangle$. $2\vec{u} - 3\vec{w} = \langle 6 - (-3), -4 - 12 \rangle = \langle 9, -16 \rangle$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Vector operations review** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

$\vec{u} \cdot \vec{w} = 3(-1) + (-2)(4) = -3 - 8 = -11$. Since the dot product is not 0, the vectors are NOT perpendicular.

16 **[Challenging]** • **Target: Curvature-related speed problems** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

$\vec{v}(t) = \langle 3t^2, 2t \rangle$. Speed$^2 = 9t^4 + 4t^2 = 13$. Let $w = t^2$: $9w^2 + 4w - 13 = 0$. Using the quadratic formula: $w = \frac{-4 \pm \sqrt{16 + 468}}{18} = \frac{-4 \pm 22}{18}$. Taking the positive root: $w = 1$, so $t^2 = 1 \Rightarrow t = \pm 1$.

17 **[Challenging]** • **Target: Curvature-related speed problems** • **Cross-Lesson: 1.01 Integration by Parts**

For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

$\vec{v}(t) = \left\langle 1, \frac{1}{t} \right\rangle$. At $t = 1$: $\vec{v}(1) = \langle 1, 1 \rangle$. Speed $= \sqrt{1^2 + 1^2} = \sqrt{2}$.

18 **[Challenging]** • **Target: Projectile motion as a vector-valued function** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

$\vec{v}(0) = \langle 20, 30 \rangle$. Speed $= \sqrt{20^2 + 30^2} = \sqrt{400 + 900} = \sqrt{1300} \approx 36.1$ m/s. The horizontal component (20 m/s) stays constant throughout the flight, while the vertical component (30 m/s initially) decreases due to gravity.

19 **[Challenging]** • **Target: Finding when a particle is at rest** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

A particle moves along $\vec{r}(t) = \langle t^3 - 3t, t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

$\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$. Need $3t^2 - 3 = 0$ AND $2t = 0$ simultaneously. From the second equation, $t = 0$; check the first: $3(0)^2 - 3 = -3 \neq 0$. There is NO time when both components are zero at once, so the particle is never fully at rest.

20 **[Challenging]** • **Target: Finding when a particle is at rest** • **Cross-Lesson: 3.01 Parametric Equations**

For the same particle, find the times when only the horizontal velocity component is 0.

$3t^2 - 3 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Position, velocity, and acceleration vectors

Given $\vec{u} = \langle 7, -1 \rangle$ and $\vec{w} = \langle 3, 8 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 23, 22 \rangle$ ✓

b. $\langle 24, 22 \rangle$

c. $\langle 27, 13 \rangle$

d. $\langle 23, -22 \rangle$

$2\vec{u} = \langle 14, -2 \rangle$; $3\vec{w} = \langle 9, 24 \rangle$. Sum: $\langle 23, 22 \rangle$.

22 [Foundational] • Target: Integrating vector-valued functions

Given $\vec{u} = \langle 7, 7 \rangle$ and $\vec{w} = \langle 2, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle -7, 10 \rangle$

b. $\langle 17, 5 \rangle$

c. $\langle -8, -10 \rangle$

d. $\langle -8, 10 \rangle$ ✓

$-2\vec{u} = \langle -14, -14 \rangle$; $3\vec{w} = \langle 6, 24 \rangle$. Sum: $\langle -8, 10 \rangle$.

23 [Foundational] • Target: Total distance traveled

Given $\vec{u} = \langle 7, 3 \rangle$ and $\vec{w} = \langle 0, 6 \rangle$, find $2\vec{u} - 2\vec{w}$.

a. $\langle 14, -6 \rangle$ ✓

b. $\langle 14, 6 \rangle$

c. $\langle 15, -6 \rangle$

d. $\langle -14, 6 \rangle$

$2\vec{u} = \langle 14, 6 \rangle$; $-2\vec{w} = \langle 0, -12 \rangle$. Sum: $\langle 14, -6 \rangle$.

24 [Foundational] • Target: Vectors and parametric motion combined

Given $\vec{u} = \langle -7, -2 \rangle$ and $\vec{w} = \langle 8, -3 \rangle$, find $3\vec{u} - 2\vec{w}$.

a. $\langle -37, 0 \rangle$ ✓

b. $\langle -39, 1 \rangle$

c. $\langle -36, 0 \rangle$

d. $\langle 38, -5 \rangle$

$3\vec{u} = \langle -21, -6 \rangle$; $-2\vec{w} = \langle -16, 6 \rangle$. Sum: $\langle -37, 0 \rangle$.

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25 [Foundational] • Target: Unit tangent vectors

Given $\vec{u} = \langle 6, -7 \rangle$ and $\vec{w} = \langle -7, -7 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle -9, 35 \rangle$

b. $\langle -9, -35 \rangle$ ✓

c. $\langle -8, -35 \rangle$

d. $\langle 4, -35 \rangle$

$2\vec{u} = \langle 12, -14 \rangle$; $3\vec{w} = \langle -21, -21 \rangle$. Sum: $\langle -9, -35 \rangle$.

26 [Foundational] • Target: Motion on a circle and centripetal acceleration

Given $\vec{u} = \langle 1, 3 \rangle$ and $\vec{w} = \langle 5, 4 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 17, -18 \rangle$

b. $\langle 13, 17 \rangle$

c. $\langle 18, 18 \rangle$

d. $\langle 17, 18 \rangle$ ✓

$2\vec{u} = \langle 2, 6 \rangle$; $3\vec{w} = \langle 15, 12 \rangle$. Sum: $\langle 17, 18 \rangle$.

27 [Foundational] • Target: Displacement vs. distance for vector motion

Given $\vec{u} = \langle 8, 0 \rangle$ and $\vec{w} = \langle 3, -1 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle -18, -2 \rangle$

b. $\langle 7, -3 \rangle$

c. $\langle 7, 3 \rangle$ ✓

d. $\langle 8, 3 \rangle$

$2\vec{u} = \langle 16, 0 \rangle$; $-3\vec{w} = \langle -9, 3 \rangle$. Sum: $\langle 7, 3 \rangle$.

28 [Foundational] • Target: Vector operations review

A particle moves along $\vec{r}(t) = \langle 5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 26 ✓

b. 28

c. 34

d. 13

$\vec{a}(t) = \langle 10, 24 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{10^2 + 24^2} = \sqrt{676} = 26$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Curvature-related speed problems

A particle moves along $\vec{r}(t) = \langle -8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 36

b. 34 ✓

c. 46

d. 17

$\vec{a}(t) = \langle -16, 30 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{(-16)^2 + 30^2} = \sqrt{1156} = 34$.

30 [Intermediate] • Target: Projectile motion as a vector-valued function

A particle moves along $\vec{r}(t) = \langle -24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 62

b. 50 ✓

c. 52

d. 25

$\vec{a}(t) = \langle -48, 14 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{(-48)^2 + 14^2} = \sqrt{2500} = 50$.

31 [Intermediate] • Target: Finding when a particle is at rest

A particle moves along $\vec{r}(t) = \langle -9t^2, -40t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 98

b. 84

c. 82 ✓

d. 41

$\vec{a}(t) = \langle -18, -80 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{(-18)^2 + (-80)^2} = \sqrt{6724} = 82$.

32 [Intermediate] • Target: Vector-valued functions and arc length

A particle moves along $\vec{r}(t) = \langle 6t^2, -8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 22

b. 20 ✓

c. 28

d. 10

$\vec{a}(t) = \langle 12, -16 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{12^2 + (-16)^2} = \sqrt{400} = 20$.

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33 [Intermediate] • Target: Position, Velocity, and Acceleration Vectors

A particle moves along $\vec{r}(t) = \langle 7t^2, 24t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 62

b. 25

c. 52

d. 50 ✓

$\vec{a}(t) = \langle 14, 48 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{14^2 + 48^2} = \sqrt{2500} = 50$.

34 [Intermediate] • Target: Finding Position from Velocity & Acceleration

A particle moves along $\vec{r}(t) = \langle -5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 13

b. 26 ✓

c. 34

d. 28

$\vec{a}(t) = \langle -10, 24 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{(-10)^2 + 24^2} = \sqrt{676} = 26$.

35 [Intermediate] • Target: Speed, Direction, and Displacement

A particle moves along $\vec{r}(t) = \langle t^3 - 108t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 6 ✓

b. 18

c. 7

d. -6

$\vec{v}(t) = \langle 3t^2 - 108, 2t \rangle$. Setting $3t^2 - 108 = 0$: $t^2 = 36 \Rightarrow t = \pm 6$. The positive value is 6.

36 [Intermediate] • Target: Distance Traveled with Vector-Valued Functions

A particle moves along $\vec{r}(t) = \langle t^3 - 75t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. 5 ✓

b. 6

c. -5

d. 15

$\vec{v}(t) = \langle 3t^2 - 75, 2t \rangle$. Setting $3t^2 - 75 = 0$: $t^2 = 25 \Rightarrow t = \pm 5$. The positive value is 5.

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37 [Intermediate] • Target: Position, velocity, and acceleration vectors

A particle moves along $\vec{r}(t) = \langle 4t, 6t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 1 ✓

b. 2

c. 3

d. 4

$\vec{v}(t) = \langle 4, 6 - 6t \rangle$. Setting $6 - 6t = 0$: $t = \frac{-(6)}{-6} = 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Integrating vector-valued functions • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves along $\vec{r}(t) = \langle 6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 12

b. 7

c. 5

d. 6 ✓

$\vec{v}(t) = \langle 6, 36 - 6t \rangle$. Setting $36 - 6t = 0$: $t = \frac{-(36)}{-6} = 6$.

39 [Challenging] • Target: Total distance traveled • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 7]$.

a. 119 ✓

b. 136

c. 161

d. 126

$\vec{v}(t) = \langle 8, 15 \rangle$, so $|\vec{v}(t)| = \sqrt{8^2 + 15^2} = 17$ (constant speed). $L = \int_0^7 17\,dt = 17(7) = 119$.

40 [Challenging] • Target: Vectors and parametric motion combined • Cross-Lesson: 3.01 Parametric Equations

Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 4]$.

a. 164 ✓

b. 205

c. 196

d. 168

$\vec{v}(t) = \langle 9, 40 \rangle$, so $|\vec{v}(t)| = \sqrt{9^2 + 40^2} = 41$ (constant speed). $L = \int_0^4 41\,dt = 41(4) = 164$.

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41 [Challenging] • Target: Position, Velocity, and Acceleration Vectors • Cross-Lesson: 4.01 Polar Curves and Derivatives

A particle has position vector $\mathbf{r}(t) = \langle t^3 - 2t, \sin(2t) \rangle$. Find the acceleration vector $\mathbf{a}(t)$ at $t = 0$.

a. $\langle 0, 0 \rangle$ ✓

b. $\langle -2, 2 \rangle$

c. $\langle 0, -4 \rangle$

d. $\langle 6, 0 \rangle$

$\mathbf{v}(t) = \langle 3t^2 - 2, 2\cos(2t) \rangle$. $\mathbf{a}(t) = \langle 6t, -4\sin(2t) \rangle$. At $t = 0$: $\mathbf{a}(0) = \langle 0, 0 \rangle$.

42 [Challenging] • Target: Position, Velocity, and Acceleration Vectors • Cross-Lesson: 4.02 Area in Polar Coordinates

What is the magnitude of the acceleration vector $\mathbf{a}(t) = \langle 3, -4 \rangle$?

a. 5 ✓

b. 7

c. $\sqrt{7}$

d. 1

$||\mathbf{a}|| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$.

43 [Challenging] • Target: Finding Position from Velocity & Acceleration • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

If a particle has $\mathbf{v}(t) = \langle \frac{1}{t+1}, 2e^{2t} \rangle$ and $\mathbf{r}(0) = \langle 3, 4 \rangle$, find the x-component $x(t)$.

a. $x(t) = \ln(t + 1) + 3$ ✓

b. $x(t) = \ln(t + 1)$

c. $x(t) = -\frac{1}{(t+1)^2} + 3$

d. $x(t) = e^{t+1} + 2$

$x(t) = \int \frac{1}{t+1} dt = \ln(t + 1) + C$. Since $x(0) = 3$, $\ln(1) + C = 3 \Rightarrow C = 3$.

44 [Challenging] • Target: Speed, Direction, and Displacement • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Which vector represents the net displacement of a particle from $t = 1$ to $t = 4$?

a. $\int_1^4 \mathbf{v}(t) dt$ ✓

b. $\int_1^4 ||\mathbf{v}(t)|| dt$

c. $\mathbf{v}(4) - \mathbf{v}(1)$

d. $\mathbf{a}(4) - \mathbf{a}(1)$

Displacement vector is $\Delta\mathbf{r} = \mathbf{r}(4) - \mathbf{r}(1) = \int_1^4 \mathbf{v}(t) dt$.

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45 **[Challenging]** • **Target: Distance Traveled with Vector-Valued Functions** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Write an integral for the total distance traveled by a particle from $t = 0$ to $t = 5$ with $\mathbf{v}(t) = \langle \cos(t^2), e^{\sqrt{t}} \rangle$.

a. $\int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$ ✓

b. $\int_0^5 (\cos(t^2) + e^{\sqrt{t}}) dt$

c. $\sqrt{\int_0^5 \cos^2(t^2) dt + \int_0^5 e^{2\sqrt{t}} dt}$

d. $\int_0^5 ||\mathbf{a}(t)|| dt$

Distance traveled is $\int_0^5 ||\mathbf{v}(t)|| dt = \int_0^5 \sqrt{(\cos(t^2))^2 + (e^{\sqrt{t}})^2}\, dt = \int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$.

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